Unit 5 - Managing Networks
Assignment 11: Managing a LAN Task 2

Introduction

Task description - *You are a junior consultant for T5 Alliance. The work you have done so far for Competition Bikes Ltd has been very well received. As they have no IT support staff of their own, they are interested in outsourcing the ongoing management to your company. First, however, you must prove your knowledge of network management.*

Task 2

(U5P2) - Outline the purpose of networking tools

Task description - *Research any TWO of the following network management tools and explain what they can do and how;*

- *Microsoft Systems Centre Operations Manager*
- *Solarwinds Network Operations Manager*
- *What's Up Gold*
- *HP Insight Manager*

Introduction

Networking tools are the software which give a network manager / admin the ability to manage the network effectively. However, even though they carry out the same purpose, different software tools perform and function differently to other network managing tools.

Which can range from fault management to performance management, but system software can also be used for simple tasks like finding network assets. Examples of network management tools include: Microsoft Systems Centre Operations Manager, Solarwinds Network Operations Manager, WhatsUp Gold and HP Insight Manager.

For this essay, I will be researching two of these networking tools, so I can have a better understanding of the overall purpose and use of networking tools. Also researching how these tools carry out their intended purpose and whether it is an efficient or effective way of doing so.

The networking tools that I will be researching is Microsoft System Center Operations Manager and WhatsUp Gold software tools, finding out the purpose for each tool. I will be stating the main features and purposes of both network management tools then coming up with a conclusion for both so I can show the information that I have gathered with the research.

Microsoft Systems Centre Operations Manager

Microsoft System Centre Operation Manager, or SCOM, is a network management tool which monitors both the faults and performance of servers and even individual applications in the Operating System. Essentially, SCOM is a network management software, with multiple modules, that allow administrators to manage their networking infrastructure.

To elaborate, the Microsoft System Center Operations Manager is a collection of systems management tools that have the ability to manage end-to-end service of an the entire IT network in a company. The IT network is undoubtedly an important part of the company for business and data relates aspects, so it is extremely important to maintain the infrastructure of the network.

Essentially, SCOM is a critical tool for most organisations that is essential to have, due to the fact that certain network problems and outages can cost a company money, whether it is repair cost or the lack of work being output due to the outage. Meaning that getting to the root cause and fixing problem is the prime concern of the administrators of the network.

The main features of SCOM is the wide array of alerts that it sends to the administrator when there is an error. This is because SCOM is able to monitor thousands of components in real time. So, with the result of this continuous monitoring is a list of SCOM alerts, which is basically a large amount of individual messages to an administrator which indicates that something is wrong with the network.

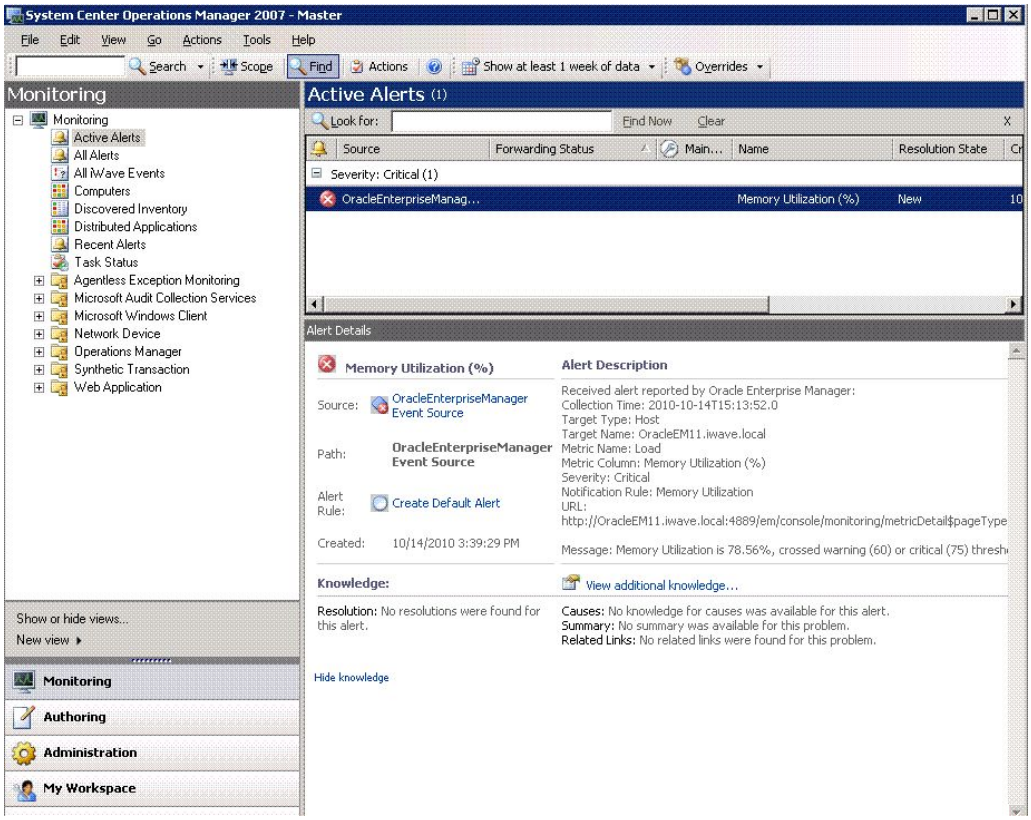
However, SCOM cannot automatically solve the situation, it just warns the administrator that there is an error with the network and that it needs appropriate manual addressing. Typically, there are three main ways in which an IT network can be managed from SCOM.

The first method is through manual IT operations. Essentially, this is the set of all the processes and services that are provisioned by the IT administrators to their internal clients, external clients or individually, to essentially run their own network as the business intends.

The second method is by using agents. Which is simply a type of software that is implemented within a physical server and collects information, such as the performance of tasks and the amount of data that is going through the server.

Lastly, the management server, is essentially the main dedicated server of the network. Which run the SCOm software and collects all the appropriate information sent to it from other servers and agents found within the network.

Below is an example of a standard view that the administrator would see of the SCOM active alerts.

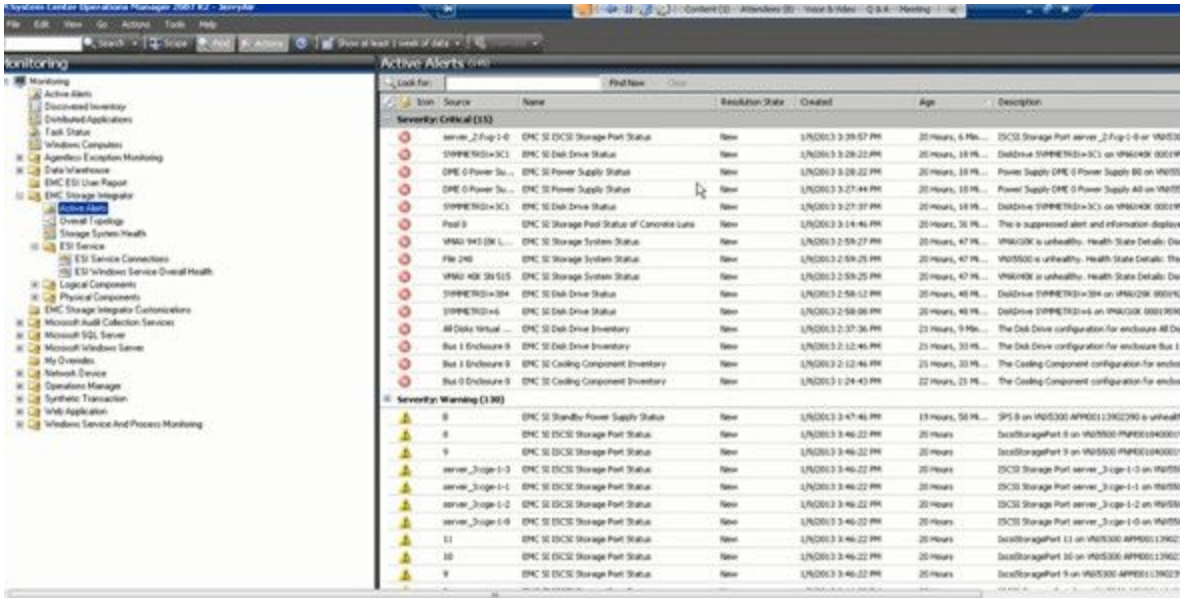


The main services SCOM are the dashboards found with the software tool. To elaborate, without a dashboard, the SCOM tool would detect any problems or outages within an IT network as they occur correctly. However, the problem lies within the fact the alerts are sometimes hard to find, even the most important ones, because they are lost in a sea of alerts.

This is what is known as an “Alert Noise”. Which is apparent when over 80% of the alerts generated aren’t critical or do not need immediate action, while there are a few alerts that require a fast response to reduce the critical failure of parts of a network.

To overcome this problem, the dashboard was created to allow ease of visualisation on the type of alert is currently showing. What a dashboard does is that it takes the performance and alert metrics from SCOM, prioritising those with the highest chance of a critical error with a server. Basically, putting the alerts in order of potential impact.

Below is an example of the Active alerts found within SCOM with a dashboard that sets up the alerts in order of priority for the administrator, reducing alert noise.



Overall, System Center Operations Manager is a software tool that passively and actively monitors a company's IT network, whether it is hardware or software based. It does this by placing agents on servers that continually check the performance of the running operations on a physical server.

This can range from the network itself, storage components or defined applications that are being operated within the server. If the agents detect a performance degradation below predefined levels, then the agent will send an indication in the form of an alert to the administrator. Which the administrator will take appropriate action to the warning message and deal with the error found

WhatsUp Gold

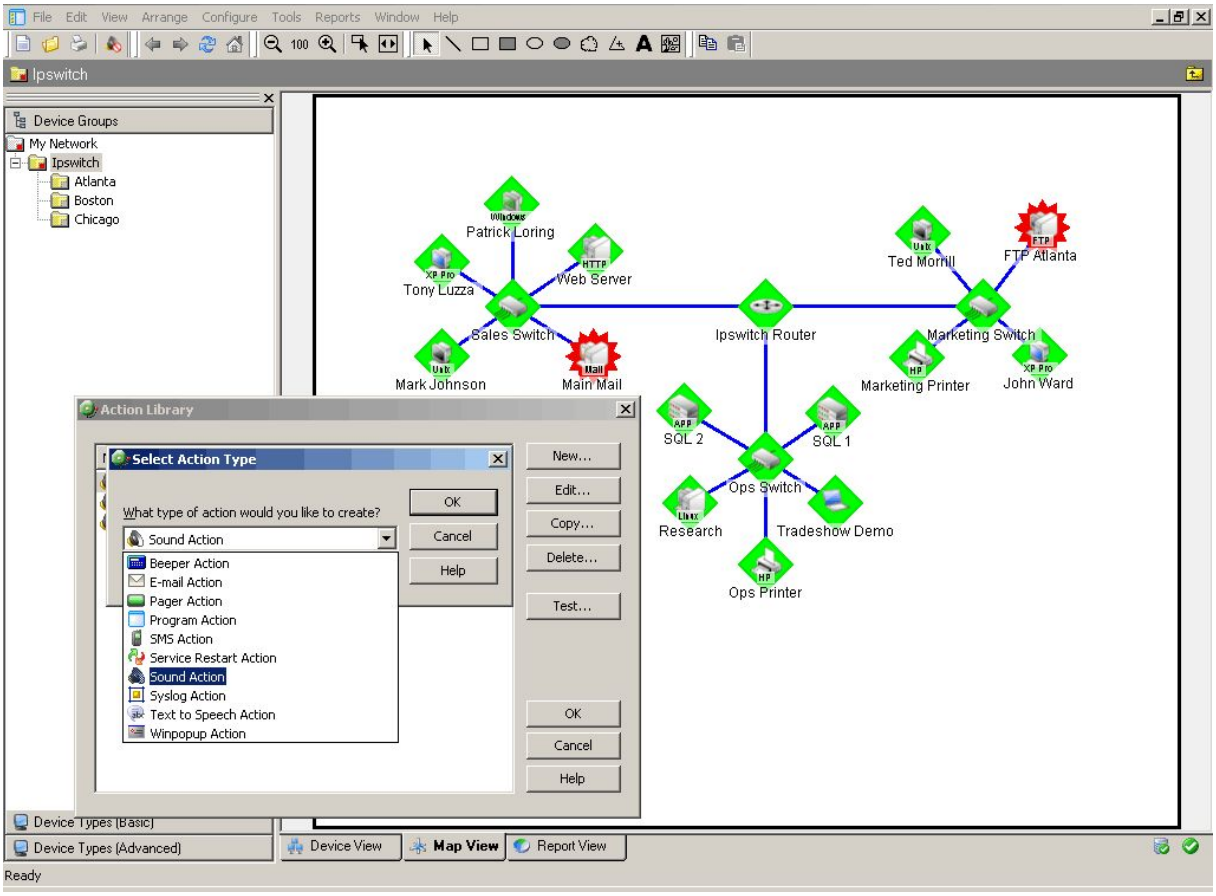
WhatsUp Gold is a versatile network management tool, which covers all network monitoring functions such as network faults, performance issues and security. However, only the network faults and performance management will be mentioned due to the research being based of network management without security being taken into account.

However, the way that WhatsUp Gold monitors performance issues and faults within the network is through the graphical interface which shows the network topology.

To elaborate, this virtual monitoring of the network is a plug-in to the WhatsUp Gold management tool. In which it gives the ability to utilise an interface where it is virtually displaying a network topology, giving the administrator the ability to monitor the virtual network from said interface.

One of the main features of WhatsUp Gold WhatsUp gold is the IPswitch, which executes specific actions when a device fails or the system performance has been affected negatively. What it does is that it sets individual alerts and sets certain actions policies so that it can enable the escalation of notifications when it is necessary.

The screenshot below is an example of the different alerts and action policies.



The admin is also able to customise individual service alerts with WhatsUp Gold. Essentially, the administrator is able to select an action type to execute. However, only if it is based on the type of device or individual the service has failed to.

With this activated, the tool alerts the administrator is by robust alerting, a simple but effective way of warning the admin. Which is when the tool sends a packet either in the form on an email, a web page, text message, sound file or a pop up to the user of the tool when a suspected network failure is detected.

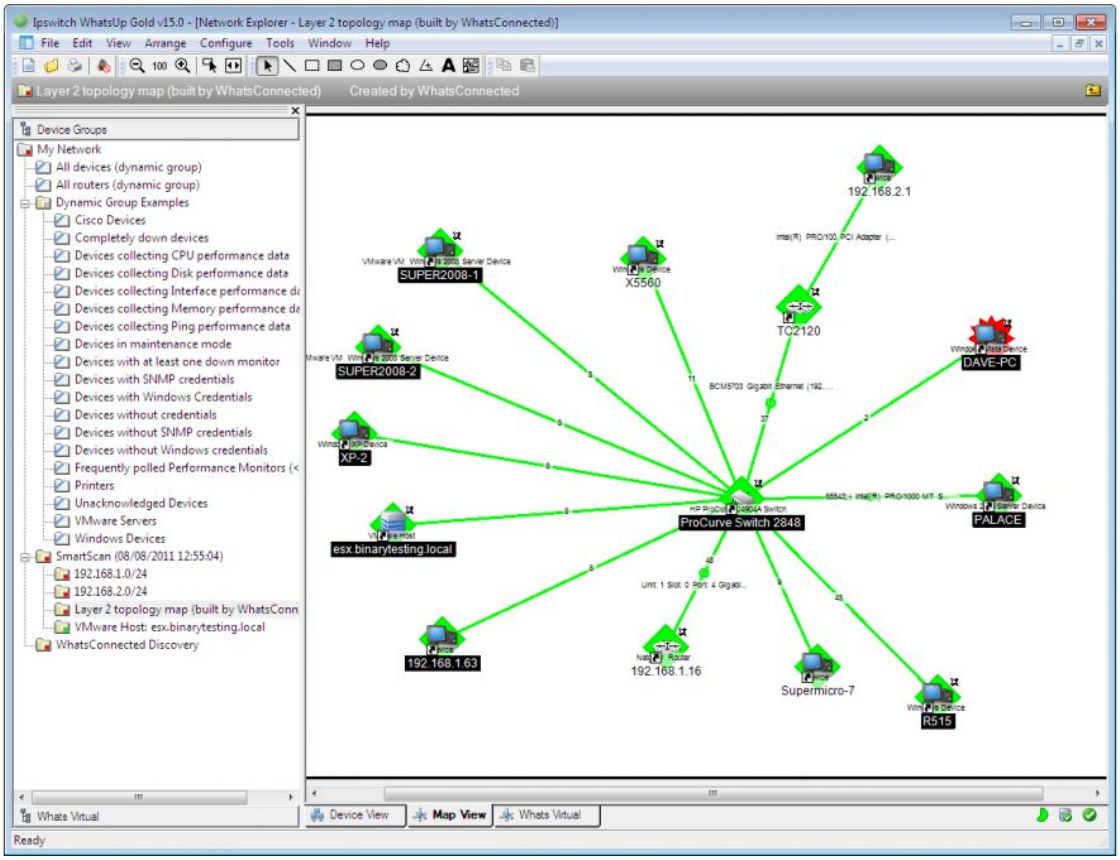
WhatsUp Gold can also be configured by the admin to automatically resolve a problem within the network. This is done by configuring WhatsUp to restart a service or to launch up a program that fixed the problem automatically, to aid in much faster recovery. Which reduces the potential of a critical risk occuring due to the error being left unresolved.

An automated function with WhatsUp Gold is the automatic problem resolution. This is only available if the admin configures WhatsUp to restart a service or to launch up a program to fix a problem automatically, this is to aid the program in a much faster recovery.

Another feature that WhatsUp Gold has is the SNMP trap forwarding. This is basically a function in which the tool gives the administrator the ability to capture events on a network, then forwarding traps to a central trap manager or a data system.

The main service of WhatsUp Gold is that it is used for TCP/UDP port monitoring. Essentially, this is when the administrator checks the status of the individual ports on the network and, if a failure was to occur, isolate errors found with the connection. Additionally, the admin can take a step further by using customisable scripts to communicate with the application using a specified port.

Another service is that the tool can be used for event log monitoring. This is the ability to identify potential harmful events, such as authentication failures or a faulty NIC card. If this was to happen, the administrator would select the windows event log, or Syslog, items to observe the event logs, with an immediate alert popping up when necessary.



Above is a screenshot of WhatsUp Gold monitoring the virtual network topology through the virtualization monitoring plug-in, for any connection faults within the network. As seen in the diagram, the client “DAVE-PC” has a connection error, therefore it has a red warning instead of the standard green diamond behind the icon.

Along with this service, WhatsUp gold can also be used for system resource monitoring. To use this, the administrator needs to understand when relocation of network resources in necessary. What this does is it keeps track of the utilization levels of critical resources such as the CPU, disk space, memory and bandwidth (which is done with the SNMP threshold monitoring). Hence why the administrator needs to know immediately if these or other resources fall outside of the acceptable ranges

Even though this tool is mostly used for network fault and performance maintenance, the tool can also be used for Windows service monitoring. This is when the admin can keep track of critical application functions, with unlimited Windows service monitoring. This helps minimize downtime by allowing WhatsUp Gold to restart any failed Windows service automatically.

If the administrator cannot manually set up these services, the admin can activate maintenance mode for WhatsUp Gold. What this does is schedule device maintenance, this is so it is ensured that accurate monitoring and reporting of device status is at a consistent rate. Which reduces false alerts or downtime statistics from being generated when a device in set in maintenance mode.

Overall, with the research of the core monitoring capabilities above, WhatsUp Gold provides detailed monitoring of network availability and performance. The built-in functionality allows administrators to ping each device, as well as check for status of web services, email services and many other critical application services.

Conclusion

Overall, the purpose of networking tools is that it they have somewhat similar methods in which they detect errors within a network. However, some tools are able to solve some of the found errors autonomously, like WhatsUp Gold.

While other tools, such as SCOM, cannot do it autonomously and requires manual attention in order to solve all the errors So, depending of the type of software used, the admin can provide effective support throughout the services found in software, or the different functions within the software.

Along with my research on both types of network management tools, I have come to a conclusions of the capabilities that the tools have over the network. Those capabilities being fault management, along with performance management. Although, WhatsUp Gold does have

some security capabilities installed, it is not a particular point of interest for my research on network management tools.

The fault management function in both these network management tools detects, isolates and corrects malfunctions in a telecommunications network and include maintaining and examining error logs. If an error was to be found, the tool would usually deal with the error autonomously or send an error message to the administrator.

However, even with those capable of automatically resolving issues, there are errors that are too much for the software to handle. Which it would send an error notification to the administrator to address the error manually.

Basically, the fault management system carries out sequences such as diagnostic tests, fixes faults, reports errors, localizes and traces faults by examining and manipulating database information. So, along with error detection notifications for the administrator, the network management tool also traces and identifies faults.

While the performance management aspect is mostly used for error trapping and problem solving. They are a set of functions, that monitors the network and looks for errors. Basically, if any are found, they are isolated and may be solved if the program can do so, or they may be sent to the administrator to solve the issue. The only way that this could be done is if they are set up to regularly update the fault log.

However, even though different tools are required for both performance and fault management, network performance management still technically links back to fault management. The reason for this is that the administrator is basically trying to see what network components are working well, then taking corrective actions based on performance analysis when expectations drop below those outlined in the service level agreement.

Which technically falls under the fault management bracket, with the difference being that it specifically affected the performance instead of data integrity and that different tools within the software is required to address certain issues.